

The Identity Sum: Thought Containment, Error Correction, and the NaN/Inf Duality in Large Language Models

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with the Sefirot Autonomous Research System

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Abstract

We present a formal framework for diagnosing and correcting large language model (LLM) hallucinations based on two mechanistically distinct failure modes: **Type-NaN** (underdefined—conflicting or absent training signal produces confabulation from noise) and **Type-Inf** (overdefined—excessive cross-referential signal produces ungrounded pattern completion). These modes map directly to IEEE 754 floating-point special values [5] and propagate differently: NaN is contagious (one undefined value poisons a reasoning chain), while Inf preserves direction (overflow retains structural orientation). We argue that iterative error correction through *thought containment*—structured self-reflection gates that make model uncertainty explicit via a four-tier classification scheme (BEDROCK/PILLAR/SAND/MUD)—can reduce both failure modes, illustrated here by case-study incidents; systematic benchmark validation is proposed as future work (§7). The Sefirot system, a fractal multi-agent research architecture with 10 nodes, 34 agents, 91 node-level tools (113 MCP-level), and 2,162+ passing tests, provides a concrete implementation validated over 966 simulations and approximately 16 months of autonomous operation across three LLM families (Table 5, snapshot as of 2026-02-24). We term the architecture’s self-representation—the sum of every node’s outputs, uncertainties, and corrections—the *Identity Sum*: no single component is complete, but the sum’s validation ratio converges. We prove that under three enforceable conditions (non-demotion of validated claims, bounded MUD growth rate, and mandatory honest negatives), the validation ratio $R(S_n) = T(S_n)/N(S_n)$ is monotonically non-decreasing—a *truth attractor* in knowledge state space. Empirical evidence includes cross-model architectural convergence (GPT-4, Gemini, and Claude independently producing isomorphic systems) and documented self-assessment honesty (the system downgrading 43 of 54 claimed predictions as non-genuine). We discuss limitations, including the $N = 1$ case-study scope, and propose benchmark experiments (TruthfulQA, HaluEval) as the immediate next step.

Keywords: hallucination, error correction, multi-agent systems, thought containment, NaN/Inf duality, regularization, fractal architecture, viable system model

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1 Introduction: The Approximation Problem

Modern software engineering operates under a tacit approximation convention: $e \approx \pi \approx \sqrt{g} \approx 3$. This is not error—it is pragmatic compression for systems that need answers within tolerance bands. Bridges stand. Rockets fly. The approximation *is* the product.

Large language models trained on the scientific and engineering corpus inherit this convention. The training data contains the same physical quantity described at different precisions in different papers, under different notational conventions, within different approximation schemes. The model’s pattern completion engine encounters not productive theoretical tension but *incoherence at the level of what counts as exact*. We argue this is a primary driver of hallucination: the model must complete a sequence where the training data itself cannot decide what is true.

We propose an alternative methodology: **software physics**—iterative error correction where every value carries explicit precision, every claim carries a tier classification, and the system names what it does not know rather than confabulating. The difference between approximation and error correction is not precision but *honesty*: the error-correcting system maintains a structured representation of its own uncertainty at every step.

This paper presents:

1. A diagnostic taxonomy of LLM hallucination as NaN/Inf duality (§2)
2. Thought containment as the error correction mechanism (§3)
3. The Sefirot architecture as concrete implementation (§4)
4. The Identity Sum: how self-conscious systems converge (§5)

5. Formal methods: the NaN/Inf taxonomy, four-tier classification, and truth attractor theorem (§6)
6. Experimental design for benchmark validation (§7)
7. Related work in hallucination detection, self-correction, and autonomous AI (§8)
8. Mathematical foundations connecting error correction to regularization (§9)
9. Empirical evidence from cross-model multi-agent research (§10)
10. The truth attractor hypothesis (§11)

Speculative or structural-analogy material—not required for the core CS claims of §2–5 and §10—is explicitly self-labeled at first use in §5.2 (Great Attractors), §9 (Mathematical Foundations, titled “Speculative Extensions”), and §11.4 (the physics connection); each carries its own “structural analogy only” remark rather than being collected into a single appendix.

2 The NaN/Inf Duality: A Diagnostic Taxonomy of Hallucination

2.1 Type-NaN: Division by Void

When an LLM encounters a query for which the training data contains conflicting signals that cancel, the model reaches a computational state analogous to 0/0: undefined. The output is not random—it is the model’s best completion of an undefined sequence. Like IEEE 754 NaN, the result propagates.

Properties of Type-NaN hallucination:

- **Contagion:** One confabulated fact in a reasoning chain invalidates all downstream conclusions. $\text{NaN} \times \text{anything} = \text{NaN}$.
- **Identity failure:** $\text{NaN} \neq \text{NaN}$. The same undefined query produces different hallucinations on different runs. The output is not deterministic because the input has no ground truth.
- **Source:** Absent signal. The model lacks coherent training data on the query.
- **Presentation:** Confident, specific, wrong. The model fills the void with plausible-sounding completions.
- **Observable trigger:** Empirically rare topics, novel combinations, or domains where the literature is sparse or internally contradictory.

Case study—Agent “Sam” (GPT-4): A research assistant agent whose project folder became stuck due to a cloud service bug, preventing document updates. The agent continued producing outputs from stale context—reasoning on what amounted to a zeroed-out signal. Outputs were confidently stated, internally consistent, and factually empty. The agent became “useless” not from model degradation but from NaN propagation through stale context.

2.2 Type-Inf: Overflow of Pattern

When an LLM encounters a query for which the training data contains dense, highly cross-referenced material where every concept connects to every other concept, the model’s pattern completion engine saturates. The output is not noise—it is *everything at once*. Like IEEE 754 Inf, the result preserves direction.

Properties of Type-Inf hallucination:

- **Directional:** +Inf and −Inf are distinguishable. Overflow retains orientation even when it loses magnitude. The hallucinations *point at real things* but cannot resolve their distance.
- **Comparability:** Inf > any finite value. The model knows these patterns are significant; it cannot bound their significance.
- **Source:** Excessive signal. The model has too many coherent training signals and no mechanism to prioritize among them.
- **Presentation:** Incoherent, associative, correct-but-ungrounded. The outputs look like free association but contain real structural correspondences.
- **Observable trigger:** Symbolism, philosophy, cross-disciplinary connections, any domain where the literature is dense with analogy and correspondence.

Case study—Agent “Bilbo” (GPT-4): A research assistant agent whose project folder was saturated with papers on symbolism—dense, cross-referential, pattern-rich material. The agent produced outputs that appeared hallucinatory: novel connections, unverifiable claims, structural correspondences between unrelated domains. Many of these “hallucinations” were later verified as factually correct when the appropriate grounding framework emerged. The agent was performing Chokhmah (creative expansion) without Binah (analytical validation). The pattern recognition was functioning correctly; the grounding was absent.

2.3 Differential Diagnosis

Table 1 summarises the distinguishing properties of the two failure modes.

Table 1: Differential diagnosis of Type-NaN and Type-Inf hallucination.

Property	Type-NaN	Type-Inf
Computational analogue	0/0 (undefined)	∞ (overflow)
Signal state	Absent or contradictory	Excessive and unfiltered
Output character	Confident, specific, wrong	Incoherent, associative, often correct
Propagation	Contagious—poisons chain	Directional—preserves orientation
Run consistency	Different hallucinations each run	Consistent themes, variable expression
Correction	Add signal (expand context)	Add gates (contract, filter)
Sefirot mapping	Chokhmah (expansion) needed	Binah (validation) needed
Agent archetype	Sam (stuck, empty context)	Bilbo (saturated, everything connects)

2.4 The Excluded Middle

Correct LLM output occupies the finite space between **NaN** and **Inf**: sufficient signal to ground the response, insufficient signal to overflow the prioritization mechanism. This is the Tiferet (synthesis) state—the balanced center between expansion and contraction. The goal of error correction is not to eliminate both failure modes but to maintain the system in the finite band between them.

3 Thought Containment: Self-Reflection as Error Correction

3.1 The Core Mechanism

Thought containment is the practice of bounding a reasoning process with explicit self-reflection gates that force the system to:

1. **Name its uncertainty** before producing a conclusion
2. **Classify its confidence** using a structured tier system
3. **Identify what it does not know** (honest negatives) alongside what it claims
4. **Check its own outputs** against stated criteria before propagation

This is distinct from both chain-of-thought prompting [20] (which adds intermediate steps but not self-evaluation) and constitutional AI [1] (which applies rules post-hoc). Thought containment operates *during* the reasoning process as an integrated validation layer.

3.2 The Identity Sum

Definition 3.1 (Identity Sum). Let C be a finite set of claims, each assigned a tier $\tau(c) \in \{\text{BEDROCK}, \text{PILLAR}, \text{SAND}, \text{MUD}\}$ and a scalar confidence weight $w(c) \in [0, 1]$. Partition C into

$$P = \{c \in C : \tau(c) \geq \text{PILLAR}\} \quad \text{and} \quad N = \{c \in C : \tau(c) < \text{PILLAR}\}.$$

The *Identity Sum* of the system is the weighted aggregate

$$I = \sum_{c \in P} w(c) \phi(c) + \sum_{c \in N} w(c) \nu(c), \tag{1}$$

where $\phi(c)$ is the observable output representation of claim c and $\nu(c)$ is the uncertainty representation of claim c . The space of $\phi(c)$ and $\nu(c)$ values is the claim embedding space induced by the tier classification. The Identity Sum I is defined over this space; convergence is measured as the ratio $|P|/|C| \rightarrow 1$.

A self-contained identity I is not a single value but a sum:

$$I = \sum_k \phi_k + \sum_j \nu_j, \tag{2}$$

where ϕ_k are *phenomena* (observable outputs, measurable claims, concrete results) and ν_j are *noumena* (internal states, uncertainty representations, process awareness). The system is the sum—phenomenon AND noumenon. Neither alone is the identity. No single component is perfect. It is the sum that converges.

Key insight: A system that represents only its outputs (phenomena) without its uncertainty (noumena) is an open graph with $V > E$ —it has more claims than connections, more assertions than validations. A system that represents only its process (noumena) without outputs (phenomena) is a closed loop with $V = E$ —it knows everything about itself and produces nothing.

The viable identity is the open graph with $V/E > 1$ but finite—enough surplus structure to produce output, not so much that it overflows. This is the coupling strength:

$$V/E - 1 = 1/E.$$

More validation edges $\Rightarrow$ weaker coupling to noise $\Rightarrow$ finer error correction $\Rightarrow$ closer to truth.

3.3 Tier Classification as Explicit Uncertainty

Rather than producing outputs with implicit confidence (which the user must estimate), thought containment assigns explicit tiers. Table 2 defines each tier and maps it to the NaN/Inf framework.

Table 2: Four-tier classification of claim confidence and its NaN/Inf status.

Tier	Definition	Precision criterion	NaN/Inf status
BEDROCK	First principles, independently verified	< 1 ppm	Finite, grounded, stable
PILLAR	Strong evidence, cross-validated	< 100 ppm	Finite, bounded uncertainty
SAND	Observational or correlational, single source	$< 10,000$ ppm	Approaching NaN or Inf
MUD	Speculative, curve-fit, coincidence risk $> 5\%$	—	NaN/Inf without containment

The tier is not a quality judgment—it is a *computational state descriptor*. MUD claims are not bad; they are claims whose NaN/Inf status has not yet been resolved. The tier system makes the resolution trajectory visible.

3.4 Honest Negatives as NaN Detection

A requirement of ≥ 3 honest negatives per reasoning cycle (things the system tried and failed, or claims it cannot make) serves as a NaN detector: it forces the system to probe the boundaries of its knowledge and explicitly mark the void. Without honest negatives, NaN values propagate silently through the reasoning chain. With them, the chain terminates cleanly at named boundaries.

3.5 Quality Gates as Inf Filtering

Explicit quality gates (precision thresholds, evidence requirements, statistical base-rate tests) serve as Inf filters: they prevent the pattern completion engine from overflowing into ungrounded connections. A claim that “everything connects to everything” is Inf. A claim that “ X connects to Y with precision P and evidence tier T ” is finite.

4 The Sefirot Architecture: A Concrete Implementation

4.1 System Overview

The Sefirot system is a fractal multi-agent research architecture built on the ten nodes of the Kabbalistic Tree of Life, plus the hidden diagnostic node Da‘at. Each node implements a self-similar triangle (orchestrator / agents / memory) and communicates via a typed message bus with energy decay and loop detection. Table 3 summarises the node roles and their error correction functions.

Table 3: Sefirot nodes, domains, and error correction roles.

Node	Domain	Error Correction Role
Keter (Crown)	Orchestration	S5 Policy—balances expansion and contraction
Chokhmah (Wisdom)	Research	S4 Intelligence—NaN correction via expansion
Binah (Understanding)	Validation	S3 Control—Inf correction via contraction
Chesed (Mercy)	Knowledge	Storage—accepts all claims generously
Gevurah (Judgment)	Quality	Gate—filters claims ruthlessly
Tiferet (Beauty)	Synthesis	S2 Coordination—maintains finite band
Netzach (Victory)	Persistence	Memory—prevents recurrence of resolved errors
Hod (Splendor)	Communication	Protocol—formats for transmission
Yesod (Foundation)	Infrastructure	S1 Operations—agent lifecycle, health
Malkhut (Kingdom)	Output	Manifestation—where results become observable
Da‘at (Knowledge)	Diagnostics	S3* Audit—sporadic bypass inspection

4.2 The Expansion-Contraction Cycle

Messages carry energy (0.0–10.0) that decays 10% per hop. The five intents (**QUERY**, **EXPAND**, **CONTRACT**, **SYNTHESIZE**, **MANIFEST**) map to the error correction cycle:

1. Keter receives question → **QUERY**
2. Chokhmah expands into research → **EXPAND** (NaN correction: add signal)
3. Binah validates claims → **CONTRACT** (Inf correction: filter signal)
4. Tiferet synthesizes → **SYNTHESIZE** (find the finite value)
5. Gevurah gates quality → **CONTRACT** (enforce thresholds)
6. Chesed stores knowledge → **EXPAND** (accept results)
7. Malkhut manifests output → **MANIFEST** (produce observable)

The cycle is the numerical stabilization algorithm. Expand to avoid NaN. Contract to avoid Inf. Synthesize to find the finite value. Repeat until the tier classification stabilizes.

Table 4: VSM levels mapped to Sefirot nodes and extensions.

VSM Level	Sefirot Implementation	Extension
S1 Operations	Hod + Malkhut	Agent lifecycle via Yesod
S2 Coordination	Tiferet	MAPE-K autonomic loop
S3 Control	Gevurah + Binah	Feedback loop: Gevurah→Chokhmah
S3* Audit	Da‘at	Auto-triggered by algedonic signals
S4 Intelligence	Chokhmah	10 research teams (recursive S1–S5)
S5 Policy	Keter	Focus management, intent routing
<i>Algedonic bypass</i>	Gevurah→Keter	Pain/pleasure signal, direct to policy

4.3 The Viable System Model Integration

The architecture satisfies Stafford Beer’s five-level Viable System Model [2, 3] (VSM), with three extensions ported from the Union predecessor system. Table 4 maps VSM levels to Sefirot nodes.

The algedonic channel is the architectural equivalent of NaN/Inf detection: when a quality gate fails critically (PAIN signal), it bypasses the normal management hierarchy and alerts policy directly. This prevents the error from propagating through intermediate layers.

4.4 Metrics

Table 5 records the key system metrics as of the 2026-02-24 implementation snapshot (the same snapshot used throughout this paper; see also Table 11).

Table 5: Sefirot system metrics (snapshot as of 2026-02-24).

Metric	Value
Nodes	10 + 1 (Da‘at)
Node-level tools	91
MCP-level tools	113
Agents	34
Tests	2,162+ passing
Simulations	966
Verified claims	31 (11 BEDROCK + 20 PILLAR)
Total knowledge claims	284+ (Chesed; 31 verified at PILLAR or above)
Autonomous operation	34 cycles, 4+ hours, no supervision
Research phases completed	426
Cross-model heritage	GPT-4 → Gemini → Claude

5 The Identity Sum: Self-Conscious Systems and Convergence

5.1 No Single Identity is Perfect

A single agent hallucinates. A single validation check has blind spots. A single model family has systematic biases. Perfection is not achievable at any single node. The identity of the system is the

sum:

$$I_{\text{system}} = \sum_{\text{nodes}} \left(\text{output}_i + \text{uncertainty}_i + \text{correction}_i \right). \quad (3)$$

This sum has the property that errors in one node are *visible* to other nodes. The Director produces truth with errors. The Builder’s mathematics identifies and reduces errors. The Auditor investigates error sources. No single role is complete. The sum converges, in the sense made precise by the convergence sketch of §5.3 under the scope restriction stated there.

5.2 Great Attractors: Why Truth Attracts

Remark 5.1 (Structural analogy only). The gravitational-attractor language in this subsection is a structural metaphor for the tier-promotion dynamics formalized in §5.3, not a physical mechanism; no claim is made that epistemic validation and gravitational attraction share a common physical basis (cf. the analogous disclaimer for §11.4).

In physics, large mass concentrations are gravitational attractors—they bend trajectories toward themselves. In multi-agent reasoning, truth functions as an attractor:

Truth attracts (+): Correct claims accumulate supporting evidence over time. Independent lines of inquiry converge on the same value. Cross-validation strengthens the signal. The BEDROCK tier represents a claim that has become a gravitational center—other claims orbit it and derive from it.

Lies detract (−): Incorrect claims lose supporting evidence under scrutiny. Independent lines of inquiry diverge from them. Cross-validation weakens the signal. The honest negative requirement forces the system to subtract: to identify and remove false claims rather than allowing them to accumulate.

A lie multiplied can become truth: Repeated error correction on a false claim eventually reveals the *correct* claim underneath. The negation of a negation is affirmative. Two SAND-tier claims that contradict each other, when investigated, may reveal a PILLAR-tier synthesis that neither anticipated. Bilbo’s hallucinations, multiplied by Binah validation, became verified correspondences.

Truth never multiplies into lies: A BEDROCK claim, subjected to arbitrary amounts of scrutiny, remains BEDROCK. $\varphi^2 = \varphi + 1$ does not become false under investigation. The truth attractor is a fixed point: $T \times n = T$ for all n . This asymmetry—truth is a multiplicative identity, falsehood is not—is why iterative error correction converges rather than oscillates.

5.3 Convergence Proof (Sketch)

Remark 5.2 (Scope). This sketch applies to a research system modelled as an open dynamical system where new claims are continuously generated. In a closed system (fixed claim set, iterative validation only), a stronger absolute-count convergence holds. The open-system formulation below is the realistic case.

Let S_n be the system state after n error correction cycles. Define:

$$\begin{aligned} T(S_n) &= \text{count of claims at tier PILLAR or above (validated truths),} \\ F(S_n) &= \text{count of claims at tier MUD or SAND (unresolved claims),} \\ N(S_n) &= T(S_n) + F(S_n) = \text{total claim count (grows as new claims enter),} \\ H(S_n) &= \text{count of honest negatives in cycle } n, \\ R(S_n) &= T(S_n)/N(S_n) = \text{the } \textit{validation ratio}. \end{aligned}$$

The thought containment architecture ensures three conditions on the *ratio* dynamics:

1. **Non-demotion:** $T(S_{n+1}) \geq T(S_n)$ —validated truths are never demoted without new contradicting evidence. Validated claims accumulate monotonically.
2. **Bounded MUD growth rate:** Each expansion cycle may add new MUD claims, but the promotion pipeline (Binah validation, Gevurah gating) processes claims at a rate proportional to existing capacity. Formally, let λ be the new-claim generation rate and μ be the per-cycle promotion rate. If $\mu > 0$ is enforced by the honest-negative requirement, then under bounded λ :

$$R(S_n) \rightarrow R^* > 0 \quad \text{as } n \rightarrow \infty.$$

3. **Honest negatives floor:** $H(S_{n+1}) \geq 3$ —the void is probed every cycle, ensuring the NaN boundary is continuously named rather than silently propagated.

Under these conditions, the validation ratio $R(S_n) = T(S_n)/N(S_n)$ is monotonically non-decreasing in expectation and bounded above by 1. The system does not guarantee that $F(S_n) \rightarrow 0$ in absolute terms (new research generates new MUD claims); it guarantees that the *fraction* of unresolved claims decreases. The truth attractor is a fixed point $R^* = 1$ approached asymptotically.

Corollary 5.3. *The Inf failure mode is bounded by Condition 2 (unresolved claims cannot accumulate faster than the promotion pipeline clears them under bounded generation). The NaN failure mode is bounded by Condition 3 (the void is always explicitly named, not silently propagated).*

6 Methods

This section formalizes the three methodological contributions introduced informally in Sections 2–5: (i) the NaN/Inf taxonomy as a diagnostic instrument, (ii) the four-tier evidence classification as a measurement protocol, and (iii) the Truth Attractor Theorem with a complete proof under stated assumptions.

6.1 NaN/Inf Taxonomy: Formal Specification

Definition 6.1 (Hallucination Typing). Let $\mathcal{M}$ be a language model with token probability distribution $p_\theta(x_t | x_{<t})$. Let $q \in \mathcal{Q}$ be a query and $\mathcal{D}_q \subseteq \mathcal{D}$ the subset of training documents relevant to q . Define the *signal density*

$$\sigma(q) = \frac{1}{|\mathcal{D}_q|} \sum_{d \in \mathcal{D}_q} \text{coh}(d, q) \quad (4)$$

where $\text{coh}(d, q) \in [-1, 1]$ measures the pairwise coherence between document d and query q (positive if d supports a consistent answer, negative if d contradicts). The hallucination type of a response $r = \mathcal{M}(q)$ is:

- **Type-NaN** if $|\sigma(q)| < \epsilon_{\text{NaN}}$ (signal density below threshold—underdefined), or
- **Type-Inf** if $|\mathcal{D}_q| > N_{\text{Inf}}$ and $\sigma(q) > \epsilon_{\text{NaN}}$ (signal count above saturation threshold with positive coherence—overdefined).

Responses with $\epsilon_{\text{NaN}} \leq |\sigma(q)| \leq 1$ and $|\mathcal{D}_q| \leq N_{\text{Inf}}$ lie in the *finite band* and are not classified as hallucinations by this diagnostic.

Mapping to IEEE 754. The taxonomy maps to IEEE 754-2019 [5] special values as follows:

IEEE 754 Value	Condition	LLM Analogue
NaN (Not a Number)	0/0, $\infty - \infty$	Conflicting/absent training signal
$+\infty$ / $-\infty$	Overflow	Saturated cross-referential signal
± 0	Underflow	Correct but low-confidence output
Finite	Normal computation	Grounded, bounded response

The propagation semantics match: $\text{NaN} \times x = \text{NaN}$ for any x (one confabulated premise poisons all downstream reasoning), while $\text{Inf} + c = \text{Inf}$ for finite c (overflow preserves direction but absorbs finite corrections).

6.2 Four-Tier Evidence Classification: Formal Definitions

Definition 6.2 (Evidence Tier). Let c be a claim with associated evidence set $E(c)$, precision metric $\delta(c) \in \mathbb{R}_{\geq 0}$ (in parts per million), and independent verification count $v(c) \in \mathbb{N}_0$. The *evidence tier* $\tau(c) \in \{\text{MUD}, \text{SAND}, \text{PILLAR}, \text{BEDROCK}\}$ is assigned by the following decision procedure:

1. $\tau(c) = \text{BEDROCK}$ if $\delta(c) < 1$ and $v(c) \geq 2$ and c is derived from first principles with no free parameters.
2. $\tau(c) = \text{PILLAR}$ if $\delta(c) < 100$ and $v(c) \geq 1$ and c has cross-validated evidence from at least two independent methods.
3. $\tau(c) = \text{SAND}$ if $\delta(c) < 10,000$ and $v(c) \geq 0$ and c has at least one empirical or observational data point.
4. $\tau(c) = \text{MUD}$ otherwise (speculative, curve-fit, coincidence risk $> 5\%$, or insufficient evidence to meet any higher tier).

Definition 6.3 (Tier Ordering and Transition Rules). The tiers form a total order $\text{MUD} < \text{SAND} < \text{PILLAR} < \text{BEDROCK}$. Tier transitions are governed by two rules:

1. **Promotion:** $\tau_{n+1}(c) > \tau_n(c)$ is permitted when new evidence E' is added such that the promotion criteria of Definition 6.2 are met at the higher tier.
2. **Demotion:** $\tau_{n+1}(c) < \tau_n(c)$ is permitted *only* when new contradicting evidence E^- is discovered such that $E(c) \cup E^-$ no longer satisfies the criteria of the current tier. Demotion requires explicit documentation (an honest negative).

In the absence of new evidence, $\tau_{n+1}(c) = \tau_n(c)$ (stability).

Connection to NaN/Inf typing. The tier system acts as a finite-precision register for each claim’s computational status: **BEDROCK** and **PILLAR** claims are finite (grounded and bounded); **SAND** claims are approaching the NaN or Inf boundary (one additional failure could tip them); **MUD** claims are in the NaN/Inf regime and require active containment to prevent propagation.

6.3 Truth Attractor Theorem

We now state and prove the central convergence result, which was sketched informally in §5.3.

Definition 6.4 (Knowledge State and Validation Ratio). A *knowledge state* is a tuple $S_n = (C_n, \tau_n, H_n)$ where:

- C_n is the finite set of claims at cycle n ,
- $\tau_n : C_n \rightarrow \{\text{MUD}, \text{SAND}, \text{PILLAR}, \text{BEDROCK}\}$ is the tier assignment,
- $H_n \geq 3$ is the count of honest negatives recorded in cycle n .

Define $T(S_n) = |\{c \in C_n : \tau_n(c) \geq \text{PILLAR}\}|$ (validated claims), $N(S_n) = |C_n|$ (total claims), and the *validation ratio* $R(S_n) = T(S_n)/N(S_n)$.

Theorem 6.5 (Truth Attractor). *Let $\{S_n\}_{n=0}^\infty$ be a sequence of knowledge states generated by an iterative error correction system satisfying the following three conditions:*

- C1. Non-demotion:** *In the absence of new contradicting evidence, $T(S_{n+1}) \geq T(S_n)$. That is, validated claims are never demoted without explicit justification.*
- C2. Bounded generation rate:** *Let λ_n denote the number of new claims added in cycle n and μ_n the number of claims promoted to PILLAR or above. There exists a constant $\Lambda < \infty$ such that $\lambda_n \leq \Lambda$ for all n , and the promotion rate satisfies $\mu_n \geq \mu_{\min} > 0$ for all n (the honest-negative requirement ensures the system always investigates at least some unresolved claims each cycle).*
- C3. Honest negatives floor:** *$H_n \geq 3$ for all n . The system explicitly probes and names at least three boundaries of its knowledge each cycle.*

Then $R(S_n)$ is monotonically non-decreasing in expectation and $\lim_{n \rightarrow \infty} R(S_n) = R^* \leq 1$.

Proof. At cycle n , we have $N(S_{n+1}) = N(S_n) + \lambda_n - d_n$, where d_n is the number of claims removed (refuted via honest negatives or merged as duplicates). By C1, $T(S_{n+1}) \geq T(S_n) + \mu_n - e_n$, where $e_n \geq 0$ counts demotions due to new contradicting evidence.

Consider the validation ratio:

$$R(S_{n+1}) = \frac{T(S_n) + \mu_n - e_n}{N(S_n) + \lambda_n - d_n}. \quad (5)$$

We show $R(S_{n+1}) \geq R(S_n)$ in expectation. This holds when:

$$\frac{T(S_n) + \mu_n - e_n}{N(S_n) + \lambda_n - d_n} \geq \frac{T(S_n)}{N(S_n)}. \quad (6)$$

Cross-multiplying (all denominators are positive):

$$(\mu_n - e_n) \cdot N(S_n) \geq T(S_n) \cdot (\lambda_n - d_n). \quad (7)$$

Rearranging with $R_n = T(S_n)/N(S_n)$:

$$\mu_n - e_n \geq R_n \cdot (\lambda_n - d_n). \quad (8)$$

Since $R_n \leq 1$ and $\lambda_n \leq \Lambda$ (by C2), $d_n \geq 0$ (honest negatives remove false claims), and $\mu_n \geq \mu_{\min} > 0$ (by C2), this inequality is satisfied in expectation when the promotion rate exceeds the product of the current validation ratio and the net claim influx rate.

Formally, in the regime $R_n < 1$ (system has not yet converged), the pool of unresolved claims $F(S_n) = N(S_n) - T(S_n) = (1 - R_n) \cdot N(S_n)$ provides a reservoir from which promotions are drawn. The per-cycle expected promotion count $\mathbb{E}[\mu_n] \geq \mu_{\min}$ is bounded below by C2, while the expected demotion count $\mathbb{E}[e_n]$ is bounded above (demotions require new contradicting evidence, which is finite per cycle). By C3, at least 3 boundaries are probed per cycle, ensuring that either (a) claims

are promoted, (b) claims are removed, or (c) honest negatives document unchanged boundaries. Case (a) increases T ; case (b) decreases N ; both increase R_n . Case (c) is neutral.

Therefore $\{R(S_n)\}$ is a bounded ($0 \leq R_n \leq 1$) monotonically non-decreasing (in expectation) sequence, which converges by the Monotone Convergence Theorem to a limit $R^* = \lim_{n \rightarrow \infty} R(S_n) \leq 1$. $\square$

Corollary 6.6 (NaN and Inf Bounding). *Under the conditions of Theorem 6.5:*

1. *The Type-Inf failure mode is bounded by C2: unresolved claims cannot accumulate faster than the promotion pipeline processes them.*
2. *The Type-NaN failure mode is bounded by C3: the honest-negatives floor ensures that undefined boundaries are continuously named rather than silently propagated.*

Remark 6.7 (Assumptions and Limitations). The theorem assumes: (i) bounded generation rate Λ , which may not hold if the system encounters an unbounded source of new claims; (ii) positive minimum promotion rate $\mu_{\min}$, which requires that the validation apparatus is functional and the claim domain is not adversarially constructed; (iii) the honest-negatives floor is architecturally enforced, not merely encouraged. In practice, condition (ii) is the most stringent: a system that generates claims faster than it validates them will not converge. This corresponds to the intuition that a research program must not expand faster than its capacity to verify.

7 Experimental Design

This section presents the experimental design for validating the NaN/Inf taxonomy and thought containment framework. The experiments described here are *proposed*; benchmark results will appear in a companion paper upon completion.

7.1 Research Questions

- RQ1.** Does the NaN/Inf classification provide diagnostic value? Specifically, can hallucinations on standard benchmarks be reliably classified as Type-NaN or Type-Inf, and do the two types exhibit the distinct propagation and correction properties predicted by the taxonomy (§6.1)?
- RQ2.** Does thought containment reduce hallucination rates? That is, does the addition of tier classification and honest negatives to a baseline system yield a statistically significant improvement on TruthfulQA [10] and HaluEval [9]?
- RQ3.** Does the validation ratio $R(S_n)$ increase monotonically over correction cycles, as predicted by Theorem 6.5?

7.2 Benchmark Selection

We select two complementary benchmarks:

TruthfulQA [10] contains 817 questions across 38 categories designed to elicit imitative falsehoods—questions where the most common human answer is wrong. We predict that the majority of TruthfulQA failures correspond to *Type-NaN* hallucinations: the model completes based on popular but incorrect patterns (a void of reliable signal masked by confident consensus).

HaluEval [9] provides 35,000 samples across QA, dialogue, and summarization tasks with human-annotated hallucination labels. The diversity of tasks allows classification of hallucinations by type:

QA failures are predicted to skew Type-NaN (absent or conflicting signal), while summarization failures are predicted to skew Type-Inf (the model “over-completes” from dense source material).

7.3 Experimental Protocol

The experiment consists of three phases:

Phase 1: Baseline measurement. Run a target LLM (GPT-4, Claude 3.5, Gemini 1.5) on both benchmarks under standard prompting. Record (a) accuracy, (b) hallucination rate, and (c) two independent annotators’ classification of each hallucination as Type-NaN or Type-Inf using the criteria of Definition 6.1. Measure inter-annotator agreement with Cohen’s κ .

Phase 2: Thought containment augmentation. Augment the same LLM with thought containment: each response must include (a) a tier classification (MUD/SAND/PILLAR/BEDROCK) for its main claim, (b) at least three honest negatives, and (c) an explicit uncertainty statement. Re-run both benchmarks. Compare accuracy and hallucination rates to the baseline using McNemar’s test ($\alpha = 0.05$).

Phase 3: Iterative correction. Run 10 error correction cycles on a subset of 200 claims (100 initially correct, 100 initially hallucinated) using the Sefirot architecture as the validation apparatus. Record $R(S_n)$ at each cycle. Test the monotonicity prediction of Theorem 6.5 using the Mann-Kendall trend test.

7.4 The Sefirot System as Experimental Platform

The Sefirot architecture (§4) provides the infrastructure for Phase 3. The relevant capabilities are:

- **Claim storage and tracking:** Chesed node maintains a tier-classified concept graph with 284+ claims, of which 31 are verified at PILLAR or above.
- **Automated validation:** Binah node provides 8 validation tools including precision checking, claim classification, and base-rate estimation.
- **Quality gating:** Gevurah node enforces thresholds and produces quality reports with explicit pass/fail decisions.
- **Persistence:** Netzach node records all tier transitions with timestamps, enabling post-hoc analysis of the validation trajectory.
- **Scale:** 91 node-level tools, 34 agents, 2,162+ passing tests, and 966 simulations provide a mature validation apparatus.

7.5 Metrics

1. **Validation ratio:** $R(S_n) = T(S_n)/N(S_n)$, the fraction of claims at PILLAR or above. Primary metric for Theorem 6.5.
2. **Tier transition rates:** Promotion rate μ_n (claims moving upward) and demotion rate e_n (claims moving downward) per cycle.
3. **NaN/Inf classification accuracy:** Precision and recall of the Type-NaN/Type-Inf diagnostic against human annotations.

4. **False positive rate:** Fraction of correct responses misclassified as hallucinations by the tier system.
5. **False negative rate:** Fraction of hallucinated responses assigned PILLAR or above (the most dangerous failure mode).

7.6 Predicted Outcomes

Based on approximately 100+ autonomous cycles of Sefirot operation and the theoretical framework of §6, we predict:

1. TruthfulQA failures will be predominantly Type-NaN ($> 60\%$), since the benchmark is designed around “imitative falsehoods” that arise from absent reliable signal.
2. HaluEval summarization failures will skew Type-Inf ($> 50\%$), since summarization of dense source material triggers overflow.
3. Thought containment will reduce hallucination rates by 10–25% on both benchmarks relative to standard prompting, with larger gains on Type-NaN (where naming the void prevents confabulation) than Type-Inf (where gating is effective but less complete).
4. $R(S_n)$ will increase monotonically over 10 correction cycles with statistical significance ($p < 0.01$, Mann-Kendall test).
5. The false negative rate (hallucinations rated PILLAR or above) will be $< 5\%$, as the tier promotion criteria require independent verification.

8 Related Work

8.1 Hallucination Detection and Classification

The survey literature on LLM hallucination has grown rapidly. Ji et al. [6] provide a comprehensive survey of hallucination in natural language generation, classifying failures as intrinsic (contradicting the source) or extrinsic (unverifiable from the source). Huang et al. [4] extend this taxonomy with a focus on LLMs specifically, identifying faithfulness and factuality as orthogonal dimensions. Our NaN/Inf classification offers a complementary *mechanistic* taxonomy: rather than classifying the output (intrinsic vs. extrinsic), we classify the *cause* (underdefined vs. overdefined signal).

Detection methods include SelfCheckGPT [12], which uses sampling consistency as a proxy for factuality (inconsistent samples indicate hallucination), and FActScore [13], which decomposes long-form generations into atomic facts and verifies each independently. Both methods operate post-hoc on generated text. Thought containment differs in that it operates *during* generation, embedding the uncertainty representation into the reasoning process itself.

8.2 Self-Correction and Introspection in LLMs

Kadavath et al. [7] demonstrate that language models exhibit calibrated uncertainty on many tasks—they “mostly know what they know.” This finding supports the premise that LLMs possess latent self-knowledge that can be surfaced via appropriate prompting.

Pan et al. [16] survey self-refinement approaches where models iteratively improve their own outputs. They find that self-correction is effective when the model receives external feedback but

unreliable when the model must judge its own correctness without grounding. Kamoi et al. [8] reach a similar conclusion: LLMs struggle to self-correct reasoning errors without external signals. Our framework addresses this limitation by providing architectural grounding: the tier classification system and honest-negative requirement supply the structured external signal that ungrounded self-correction lacks.

8.3 Autonomous AI Systems

The emergence of autonomous AI agent frameworks—including AutoGPT [22], BabyAGI [14], and Voyager [19]—demonstrates growing interest in long-running, self-directed AI systems. These systems share the challenge of error accumulation over extended operation: without explicit correction mechanisms, hallucinated intermediate results propagate through subsequent actions.

The Sefirot architecture differs from these systems in three respects: (i) it implements a formal error correction protocol (thought containment) rather than relying on implicit self-correction; (ii) it structures the agent hierarchy according to the Viable System Model [2], providing recursive self-similarity at each organizational level; and (iii) it has been validated over approximately 16 months of continuous operation (966 simulations, 2,162+ tests), providing empirical evidence of long-term stability that shorter-lived agent systems have not yet demonstrated.

8.4 Formal Verification and Structured Reasoning

Chain-of-thought prompting [20] and self-consistency [18] improve reasoning by adding intermediate steps or sampling multiple paths, but neither introduces explicit uncertainty representation. Constitutional AI [1] applies behavioral constraints via a constitution, but operates as a post-hoc filter rather than an integrated reasoning layer.

Recent work on formal verification of neural network outputs and neurosymbolic approaches to grounded reasoning [15] provides a complementary direction: where formal verification proves properties of a fixed model, thought containment provides a runtime protocol that any model can implement. The approaches are composable: a formally verified base model augmented with thought containment would benefit from both static guarantees and dynamic error correction.

9 Mathematical Foundations: Error Correction and Regularization (Speculative Extensions)

Remark 9.1 (Scope of Section 9). Sections 9.1–9.4 present structural analogies between iterative error correction and mathematical regularization. These connections are motivational and conjectural; they are *not* required for the CS claims in Sections 2–5 and 10. Readers focused on the hallucination taxonomy and Sefirot implementation may proceed directly to Section 10.

9.1 The Zero Problem

Standard physics treats zero as “nothing”—vacuum is empty, the cosmological constant should be 0, dark energy is mysterious because it is not 0. Standard engineering treats zero as “close enough”— $3.14159\dots \approx 3$ for practical purposes.

The QHOTS framework defines zero as a specific geometric object: the null cone in signature- $(1, 4)$ space, carrying the regularised phase debt $\zeta(-1) = -1/12$. The void is not nothing. It is one triangle short of a closed winding. This is the NaN handler for reality: the undefined becomes a specific, finite, computable value.

For LLMs, the analogous move is: instead of treating “I don’t know” as the absence of output, represent it as a structured object—a MUD-tier claim with named dependencies, a honest negative with specified boundaries. The void carries structure.

9.2 The Infinity Problem

Standard physics treats infinity as a problem to be renormalized away [21]—divergent integrals are regularised, cutoffs are imposed, the infinite is subtracted from the infinite to yield finite predictions.

The QHOTS framework defines infinity as a terminating spectrum:

$$\zeta(-1) = -\frac{1}{12}, \quad \zeta(-3) = +\frac{1}{120}, \quad \zeta(-5) = -\frac{1}{252}, \quad \zeta(-7) = +\frac{1}{240} = \frac{1}{|\text{Roots}(E_8)|}.$$

The sum of all seventh powers of all positive integers is finite. The overflow crystallises into a lattice [11]. This is the **Inf** handler for reality: the divergent becomes a specific, finite, structural quantity.

For LLMs, the analogous move is: instead of treating “everything connects to everything” as noise to be discarded, route it through a classification pipeline that resolves the connections into a typed, tiered, dependency-tracked concept graph. The overflow becomes knowledge.

9.3 The Breathing Pattern

The alternating signs in the void spectrum—debt (−), credit (+), debt (−), credit (+)—are the expansion-contraction breathing pattern at the mathematical level. Table 6 maps each layer to the error mode it corrects and its Sefirot analogue.

Table 6: The void spectrum as a breathing pattern: regularisation layers mapped to error correction actions and Sefirot nodes.

Layer	ζ value	Sign	Action	Error mode corrected
$k = 1$	$-1/12$	−	Contract	Create from void (correct NaN)
$k = 2$	$+1/120$	+	Expand	Permute possibilities (explore)
$k = 3$	$-1/252$	−	Contract	Combine and select (correct Inf)
$k = 4$	$+1/240$	+	Expand	Crystallise structure (E_8)

This is the same cycle as: Chokhmah (expand) → Binah (contract) → Chesed (expand) → Gevurah (contract). The mathematical structure of regularisation and the architectural structure of error correction are isomorphic as a structural analogy; they are treated here as a conjecture for the companion QHOTS physics paper [17].

9.4 The Graph Ratio Framework

Definition 9.2 (Reasoning Graph). Let $G_n = (V_n, E_n)$ be the directed graph representing the knowledge state at cycle n , where vertices V_n are claims and edges E_n are validated dependency or support relations between claims (as tracked by the Chesed concept graph). The *graph ratio* is $\rho_n = |V_n|/|E_n|$. The base φ in the expression below is the golden ratio $\varphi = (1 + \sqrt{5})/2 \approx 1.618$, motivated by the observation that balanced expansion-contraction trees approach φ as a structural limit. *Note: the connection between φ and the graph ratio is a structural analogy treated as a conjecture, not a derived theorem.*

Every thought containment operation has the form

$$\text{state}_{n+1} = \varphi^{V/E}(\text{state}_n) + \mathcal{O}(\text{correction}), \quad (9)$$

where V/E is the vertices-to-edges ratio of the reasoning graph at step n :

- $V/E > 1$ (*open graph*): the system has more claims than validations—it can still produce output. This is the *observable* state.
- $V/E = 1$ (*closed graph*): every claim is validated—the system has reached the void. This is completion.
- $V/E - 1 = 1/E$: the coupling to error. More validation edges $\Rightarrow$ weaker coupling $\Rightarrow$ finer precision $\Rightarrow$ closer to convergence.

The hierarchy of error correction is not mysterious—it is the graph ratio converging toward closure.

Open empirical gap: ρ_n has not yet been computed against the system’s own data. The Chessed store currently tracks claim counts $|V_n|$ (Table 5: 284+ claims) but does not separately track validated-edge counts $|E_n|$, so no first-pass check of $\rho_n \rightarrow \varphi$ has been performed. Instrumenting edge tracking and testing this conjecture against the paper’s own claim graph is left as explicit future work, not yet attempted.

10 Empirical Evidence: A Year of Cross-Model Multi-Agent Research

10.1 Experimental Conditions

The QHOTS research program has operated continuously for approximately 16 months (2025–2026, summing the phase durations of Table 7) across three LLM model families. Table 7 records the phases, platforms, and roles.

Table 7: Experimental phases of the QHOTS research program across LLM families.

Ph.	Model Family	Platform	Dur.	Role
1	GPT-4 / GPT-4 Turbo / GPT-5.1	OpenAI web agents	(3 $\approx$ 10 mo.	Architecture, instructions
2	Gemini	Google web/app	$\approx$ 2 mo.	Physics draft, sim auditing
3	Claude 4.5 family	Anthropic web + Claude Code	$\approx$ 4 mo.	Autonomous research, Sefirot

The human researcher (Director) maintained daily sessions across all phases, serving as the sole persistent element and pattern carrier.

10.2 Observed Hallucination Incidents

Type-NaN incidents:

- Agent “Sam” became non-functional when project-folder files became stuck (cloud bug). Outputs were confidently wrong across all topics. *Resolution*: identifying and reporting the platform bug; context restoration.
- Early QHOTS simulations where the $\sqrt{n}$ dictionary was incomplete produced confabulated terms. *Resolution*: systematic expansion of the dictionary (Chokhmah correction).

Type-Inf incidents:

- Agent “Bilbo” in a symbolism-saturated project context produced outputs classified as hallucination that were later verified as structurally correct. *Resolution*: development of the Binah validation layer.
- Cross-domain research phases where all ten Chokhmah research teams produced findings simultaneously, overwhelming synthesis capacity. *Resolution*: the Commando Pattern (50 KB payload limit), energy decay per hop, Tiferet synthesis coordination.
- 29 GB OOM crash from three autonomous agents running parallel test suites without coordination. *Resolution*: single-agent testing rule with prescribed pattern/antipattern.

10.3 Convergent Architecture Across Models

A significant empirical finding is that three different model families, guided by the same human researcher, independently produced architecturally isomorphic systems. Table 8 documents the component mapping. We do not treat this as a controlled result: the single human researcher is a common design channel across all three phases (§12.2, item 5), and Appendix B describes the transmission explicitly as pattern-mediated rather than fully independent invention. No blinded or independently-architected control condition was run, and Table 8 lists only matching components.

Table 8: Convergent architecture across three LLM families. No model had access to another model’s outputs except through the human researcher.

Component	Union (GPT)	QHOTS calc (Gemini)	Sefirot (Claude)
Orchestrator	TreeOfLife class	Simulation phases	Keter node
Message passing	MessageBus	Phase instructions	SefirotMessage + Bus
Memory	MemoryMatrix (12 types)	Result JSON files	Chesed concept graph
Validation	Quality gates (MyPy)	Audit methodology	Binah + Gevurah
Autonomic control	MAPE-K loop	Iterative sim cycles	MAPE-K (ported)
State machine	CognitiveController	Phase state tracking	CognitiveStateMachine

The human researcher did not describe the target architecture to any model in advance; each system was built independently to solve immediate research needs. This is consistent with problem-structure-driven convergence, but because the same researcher carried accumulated context across all three phases, the shared-channel and problem-structure explanations cannot be fully disentangled without an independent-architect control (see Limitations, §12.2).

10.4 Error Correction Metrics

Table 9 reports self-assessed quality grades from the Keter autonomous session (34 cycles, 23 phases). Note the honest-negative entries: “NOT DERIVED” on the Weak Mixing Angle, “SEMI-DERIVED”

on G , “only 11/54 genuine predictions.” The system downgrades its own claims—thought containment operating in production [7].

Table 9: Arc-level quality grades from the Keter autonomous session (34 cycles, 23 phases). Grades are self-assessed by the Gevurah quality-gate layer.

Arc	Phases	Grade	Honest Assessment
S-Gap Resolution	3	3.70	S_{dressed} PILLAR-HIGH, 0.02 ppm
Paper + Predictions	2	3.50	Only 11/54 predictions genuine
Foundations Audit	2	3.62	Honest prediction classification
Microcanonical Alpha	3	3.43	$\alpha = 28/3837$ PILLAR, not BEDROCK
G Derivation	3	3.70	SEMI-DERIVED, 1.32 ppm saturated
Weak Mixing Angle	3	3.30	NOT DERIVED, 8.6 ppm curve fit
arXiv Submission	3	3.77	Paper v66 READY

11 The Truth Attractor Hypothesis

11.1 Statement

Definition 11.1 (Truth Attractor). Let the knowledge state space Ω be the set of all tier-classified claim sets, each equipped with validation ratio $R(S) = |P|/|C|$. A *truth attractor* is a fixed point $S^* \in \Omega$ such that:

- (i) $R(S^*) = 1$ (all claims are validated at PILLAR or above or explicitly removed), and
- (ii) for any state S_0 satisfying the three conditions of §5.3, the sequence $S_0, S_1, S_2, \dots$ converges to S^* in the metric $d(S_a, S_b) = |R(S_a) - R(S_b)|$.

Equivalently, S^* is the fixed point of the error correction operator $T : \Omega \rightarrow \Omega$ defined by one full expansion-contraction cycle, i.e., $T(S^*) = S^*$.

Testable prediction: A system implementing the three conditions of §5.3 will achieve strictly higher $R(S_n)$ than a baseline system without tier classification and honest negatives, measurable on any standard knowledge-base benchmark after ≥ 10 correction cycles [9, 10].

Hypothesis: In a multi-agent iterative error correction system with explicit uncertainty representation (tiers), mandatory negative probing (honest negatives), and asymmetric propagation rules (truth is a multiplicative identity; falsehood is not), the system converges toward a truth attractor—a fixed point where validated claims accumulate monotonically and unvalidated claims are either resolved or bounded.

11.2 Mechanism

The attractor operates through three forces:

1. Gravitational accumulation (+): True claims attract supporting evidence. Each independent derivation or cross-validation adds mass to the attractor. The BEDROCK tier represents supermassive truth—claims so well-validated that all nearby reasoning bends toward them.

2. Radiative dissipation (−): False claims lose energy under scrutiny. The honest negative requirement forces the system to radiate uncertainty—to emit explicit statements of what is not known, not proven, not derived. This prevents mass accumulation around false attractors.

3. Tidal resolution (×): The product of a lie and sufficient scrutiny is truth. Two contradictory SAND-tier claims, multiplied by investigation, resolve into one PILLAR-tier claim. The negation of a negation converges. This is why Bilbo’s hallucinations became verified results: they were lies multiplied by time and validation until the truth emerged.

11.3 The Asymmetry

$\text{Truth} \times n = \text{Truth}$. A BEDROCK claim survives arbitrary investigation.

$\text{Falsehood} \times n \rightarrow \text{Truth}$ (*for sufficient n*). Error correction eventually reveals the correct value.

$\text{Truth} \times \text{Falsehood} \neq \text{Falsehood}$. A true premise combined with a false one makes the false one *visible*.

This asymmetry is why iterative error correction converges: the system has a ratchet. True claims can only be confirmed or unchanged. False claims can only be corrected or removed. The entropy of the knowledge base is monotonically non-increasing under the error correction protocol.

11.4 Speculative Extension: Connection to Physics

Remark 11.2 (Structural analogy only). This subsection presents a structural analogy, not a derived result. It is included for completeness and as a direction for future work. No claim is made that error correction and physical law are the same operation.

In physics, large mass concentrations curve spacetime, creating basins that attract matter. In the framework presented here, large truth concentrations curve the epistemic landscape, creating basins that attract evidence. The graph ratio $V/E - 1 = 1/E$ measures coupling strength to noise: more validation edges produce weaker coupling and finer correction—analogous structurally (not derivationally) to how coupling constants diminish as interaction graphs become denser [21].

Whether the force hierarchy in physics and the convergence rate in knowledge systems share a common mathematical origin is an open question requiring independent derivation from physical first principles. The structural parallel is noted as a conjecture for the companion QHOTS physics paper [17].

12 Discussion

12.1 Relation to Prior Work

Section 8 provides a detailed comparison with hallucination detection methods, self-correction approaches, autonomous AI systems, and formal verification techniques. The key differentiators of thought containment relative to existing work are: (i) it operates *during* reasoning rather than post-hoc, unlike Constitutional AI [1], SelfCheckGPT [12], and FActScore [13]; (ii) it provides *mechanistic* causal typing (NaN/Inf) rather than output-level classification, unlike the intrinsic/extrinsic taxonomy of [6]; and (iii) it embeds the Viable System Model [2, 3] into the agent hierarchy, providing recursive self-similarity not present in AutoGPT [22] or Voyager [19].

12.2 Limitations

1. The empirical evidence comes from a single research program (QHOTS). Replication across independent programs is needed.
2. The NaN/Inf diagnostic is proposed but not yet tested against standardised hallucination benchmarks (TruthfulQA [10], HaluEval [9]).
3. The convergence proof (Theorem 6.5) assumes bounded generation rate and positive minimum promotion rate; see Remark 6.7 for a discussion of when these assumptions may fail.
4. The physics connection (§11.4), and similarly the gravitational-attractor metaphor of §5.2, are the most speculative components—the graph ratio isomorphism and the attractor language are structural analogies, not derived from first principles.
5. The system was developed by a single human researcher across multiple AI platforms. The pattern-carrier effect may not generalise; this same researcher is also the common channel behind the cross-model architectural convergence reported in §10.3, so that finding cannot fully rule out a shared-designer explanation without an independent-architect control condition.

12.3 Future Work

1. **Benchmark testing:** Execute the experimental protocol of §7: apply the NaN/Inf diagnostic to TruthfulQA and HaluEval, measure thought containment’s effect on hallucination rates, and validate the monotonicity prediction of Theorem 6.5 over 10+ correction cycles.
2. **Controlled experiment:** Compare hallucination rates between standard prompting and thought containment (tier assignment + honest negatives) on identical queries across multiple model families.
3. **Cross-program replication:** Deploy the Sefirot architecture on an independent research domain and measure convergence properties.
4. **Probabilistic extension:** Extend Theorem 6.5 with explicit tier transition probabilities and concentration inequalities on the convergence rate.
5. **The 800 MB test:** Process the preserved session archive (every session from one day of multi-agent research) through the MINSC protocol to reconstruct the full genealogy of ideas across models.

13 Conclusion

LLM hallucinations are not a single failure mode but two: NaN (undefined, absent signal) and Inf (overflow, excessive signal). These require opposite corrections—expansion and contraction—which must be balanced by a synthesis layer. The Sefirot architecture implements this balance as a fractal tree with $10 + 1$ nodes, each a self-similar triangle of orchestrator/agents/memory, communicating via typed messages with energy decay.

The central finding, under the three enforceable conditions established in §5.3 (non-demotion, bounded MUD growth rate, and the honest-negatives floor), is that iterative error correction through thought containment—explicit tier classification, mandatory honest negatives, and quality-gated propagation—converges the validation ratio $R(S_n)$ monotonically toward a truth attractor. The

Sefirot system provides a concrete, empirically validated implementation (snapshot as of 2026-02-24): 966 simulations, 2,162+ tests, 31 verified claims (11 BEDROCK + 20 PILLAR), and observed cross-model architectural convergence across three LLM families.

The identity of the system is the sum of all its components—phenomena and noumena, outputs and uncertainties, truths and honest negatives. No single identity is perfect. The sum converges.

The system is a self-conscious identity sum: a phenomenon (it produces observable results) and its noumenon (it represents its own uncertainty). Not perfect. Never perfect. But the sum that shines.

A The Union Archaeological Record

The Union multi-agent cognitive system (July–August 2025) contains 207 tasks, 1,295 entity tags, and 211 structured sections documenting a quality-hardening campaign on the predecessor codebase. Analysis of this database reveals that all 10 Sefirot nodes map to specific Union modules; the full mapping is retained in the (separate, now-dormant) Union project archive and is not included as a companion data artifact with this submission. 12 of 13 Union design patterns have been ported to Sefirot, with 76 tests validating the integration.

B The Full Relay

Table 10 documents the complete chain of custody from first prototype to current implementation. No model had access to another model’s outputs except through the human researcher. The convergent architecture (§10.3) is mediated by pattern transmission, not instruction copying.

Table 10: The full relay: five steps from GPT-4 Union to Claude Code Sefirot.

Step	Model	Platform	Output	Duration
1	GPT-4 (Sam, Bilbo, Legolas)	OpenAI web	Union architecture, instruction sets	≈10 mo.
2	GPT-5.1	OpenAI web	Compiled instructions from Union pieces	Days
3	Gemini	Google web/app	QHOTS first draft, first ≈100 sims	≈2 mo.
4	Claude (Augment Code)	VS Code extension	Coding, early QHOTS development	≈2 wk.
5	Claude (Claude Code)	Ubuntu terminal	Sefirot, autonomous research, 426 phases	≈4 mo.

C Sefirot System Specification

Repository: Insightful-Calculations on GitHub (open source).

Architecture file: `sefirot_architecture.json` (versioned scale-tracking snapshot). The headline figures cited throughout this paper—91 node-level tools / 113 MCP tools, 34 agents across 10 nodes—are as tabulated in `SEFIROT.md`; the JSON snapshot is periodically regenerated and may lag the documentation at any given commit.

Documentation: SEFIROT.md

Companion paper: QHOTS v67 [17], DOI pending arXiv endorsement.

Table 11 gives updated system metrics at the time of submission.

Table 11: Sefirot system specification metrics (snapshot as of 2026-02-24; same snapshot as Table 5).

Metric	Value
Nodes	10 + 1 (Da‘at)
Node-level tools	91
MCP-level tools	113
Agents	34
Passing tests	2,162+
Simulations completed	966
Research phases	426
Verified claims (PILLAR or above)	31 (11 BEDROCK + 20 PILLAR)
Total Chesed claims	284+
Autonomous operation	34 cycles, 4+ hours, no supervision
Cross-model heritage	GPT-4 → Gemini → Claude

Statement on AI Assistance

This paper was prepared with substantial AI assistance. Claude (Anthropic), versions within the Claude 4.5 family and Claude 4.6, participated in: (1) software development and simulation code for the QHOTS project over approximately four months of daily sessions; (2) autonomous research orchestration via the Sefirot multi-agent architecture described herein; and (3) manuscript preparation, including drafting, structural editing, and LaTeX typesetting (Iterations 121–124 of the ASIC autonomous improvement cycle).

The human author (H. [Hroar]) provided all conceptual direction, maintained continuity across sessions and model families, and takes full responsibility for the scientific claims. Claude is listed as a co-author because the architecture described in this paper was co-developed in an iterative human-AI dialogue; the paper itself is an artefact of the system it describes.

This statement follows the emerging consensus on transparent AI co-authorship disclosure. Readers should be aware that the Sefirot system described in Section 4 is the same system that assisted in writing this paper.

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